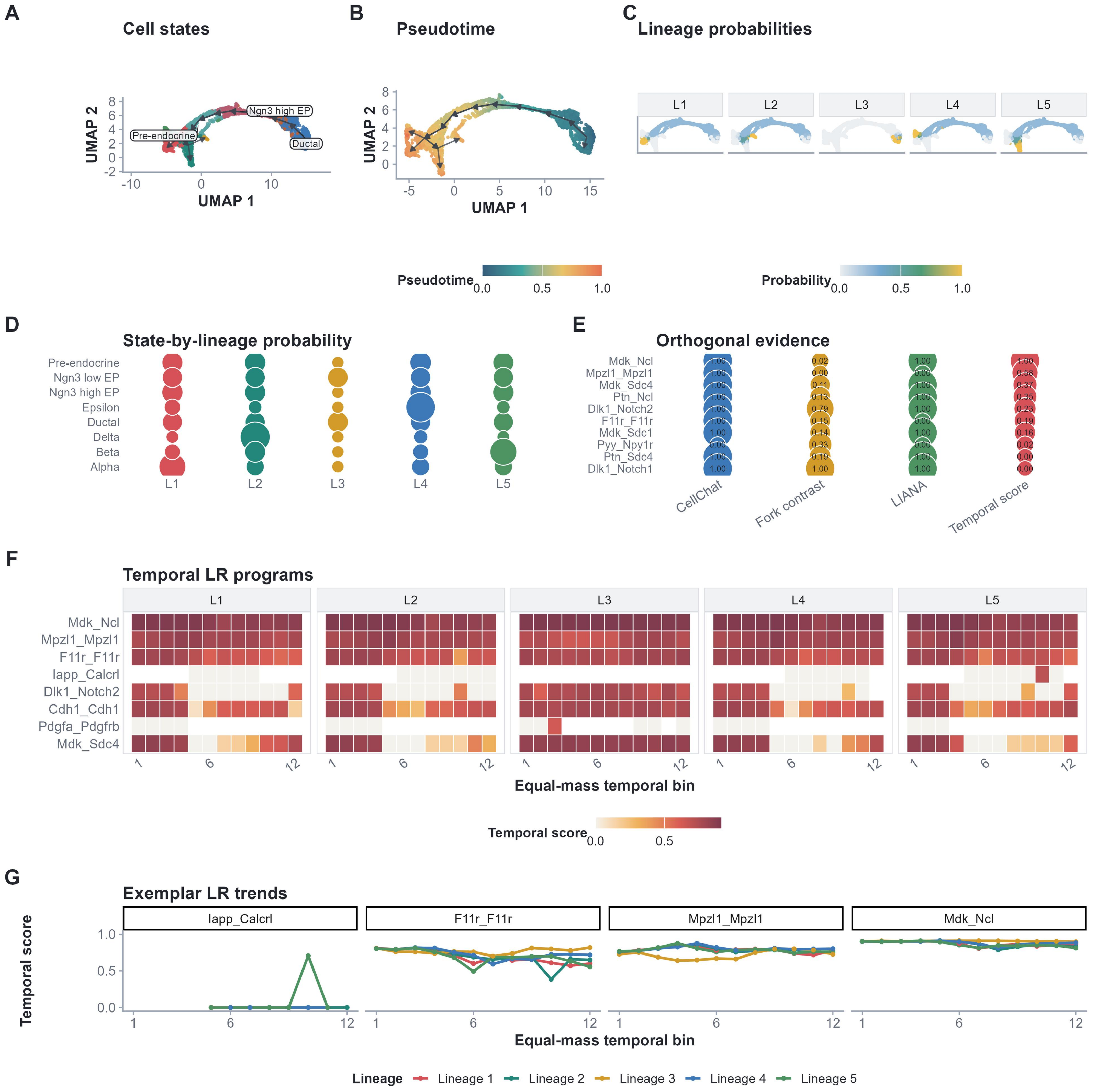
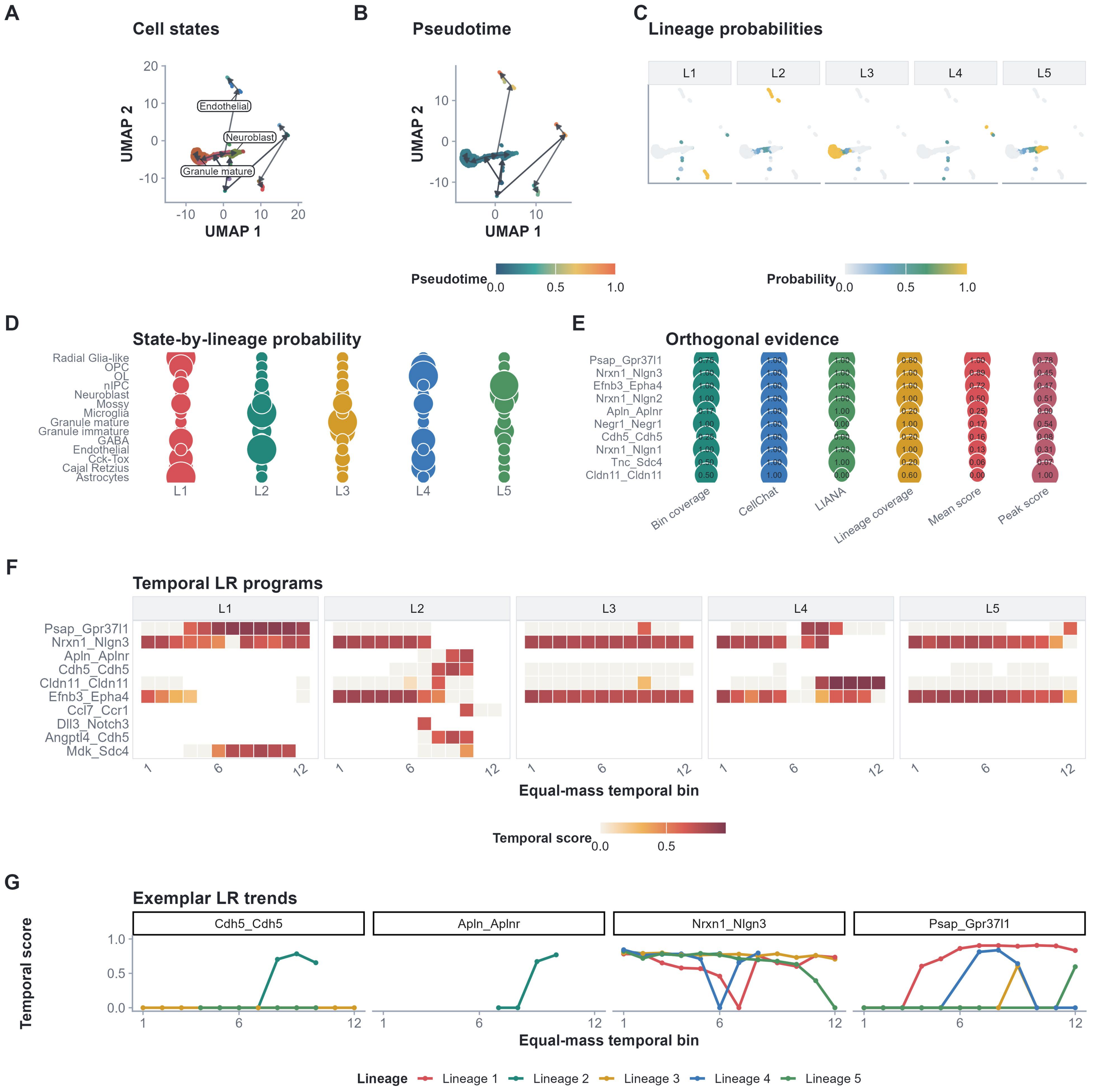

A
Cell states
UMAP 2
UMAP 1
Pre-endocrine
Ngn3 high EP
Ductal
B
Pseudotime
C
Lineage probabilities
L1
L2
L3
L4
L5
Pseudotime
0.0
0.5
1.0
Probability
0.0
0.5
1.0
D
State-by-lineage probability
Pre-endocrine
Ngn3 low EP
Ngn3 high EP
Epsilon
Ductal
Delta
Beta
Alpha
E
Orthogonal evidence
Mdk_Ncl
Mpzl1_Mpzl1
Mdk_Sdc4
Ptn_Ncl
Dlk1_Notch2
F11r_F11r
Mdk_Sdc1
Pyy_Npy1r
Ptn_Sdc4
Dlk1_Notch1
CellChat
Fork contrast
LIANA
Temporal score
F
Temporal LR programs
Mdk_Ncl
Mpzl1_Mpzl1
F11r_F11r
Iapp_Calcrl
Dlk1_Notch2
Cdh1_Cdh1
Pdgfa_Pdgfrb
Mdk_Sdc4
1
6
12
Equal-mass temporal bin
Temporal score
0.0
0.5
G
Exemplar LR trends
Iapp_Calcrl
F11r_F11r
Mpzl1_Mpzl1
Mdk_Ncl
Temporal score
1.0
0.5
0.0
1
6
12
Equal-mass temporal bin
Lineage
Lineage 1
Lineage 2
Lineage 3
Lineage 4
Lineage 5


A
Cell states
UMAP 2
UMAP 1
Endothelial
Neuroblast
Granule mature
B
Pseudotime
C
Lineage probabilities
L1
L2
L3
L4
L5
Pseudotime
0.0
0.5
1.0
Probability
0.0
0.5
1.0
D
State-by-lineage probability
Radial Glia-like
OPC
OL
nIPC
Neuroblast
Mossy
Microglia
Granule mature
Granule immature
GABA
Endothelial
Cck-Tox
Cajal Retzius
Astrocytes
E
Orthogonal evidence
Psap_Gpr37l1
Nrxn1_Nlgn3
Efnb3_Epha4
Nrxn1_Nlgn2
Apln_Aplnr
Negr1_Negr1
Cdh5_Cdh5
Nrxn1_Nlgn1
Tnc_Sdc4
Cldn11_Cldn11
Bin coverage
CellChat
LIANA
Lineage coverage
Mean score
Peak score
F
Temporal LR programs
Psap_Gpr37l1
Nrxn1_Nlgn3
Apln_Aplnr
Cdh5_Cdh5
Cldn11_Cldn11
Efnb3_Epha4
Ccl7_Ccr1
Dll3_Notch3
Angptl4_Cdh5
Mdk_Sdc4
1
6
12
Equal-mass temporal bin
Temporal score
0.0
0.5
G
Exemplar LR trends
Cdh5_Cdh5
Apln_Aplnr
Nrxn1_Nlgn3
Psap_Gpr37l1
Temporal score
1.0
0.5
0.0
1
6
12
Equal-mass temporal bin
Lineage
Lineage 1
Lineage 2
Lineage 3
Lineage 4
Lineage 5